

Multiple Choice (10 marks)

1. Occasionally a boy is born with undescended testes (cryptorchidism), and he will be sterile unless they are surgically caused to descend into the scrotum. Explain the reason for this.
 - (a) Human sperm cannot develop at the high temperature found within the body cavity.
 - (b) The testes require the lower pressure environment of the scrotum for sperm maturation
 - (c) Testes must be in the scrotum to properly attach to the vas deferens
 - (d) The scrotum provides the necessary nutrients for sperm development
 - (e) Gravity is required for the testes to function properly
2. Sea divers are aware of a danger known as the "bends". Explain the physiological mechanism of the bends.
 - (a) Carbon dioxide bubbles forming in the blood causing acidosis
 - (b) Rapid increase in barometric pressure causing the bends
 - (c) Formation of helium bubbles in the bloodstream due to rapid ascent
 - (d) Increased salinity in the water leading to dehydration and joint pain
 - (e) Oxygen bubbles forming in the blood leading to hypoxia
3. What is the function of the contractile vacuole?
 - (a) Protects the cell from foreign invaders
 - (b) Acts as a storage for genetic material
 - (c) Regulates the pH level within the cell
 - (d) Transports oxygen throughout the cell
 - (e) Removes nitrogenous wastes
4. Why is genetic drift less important in human evolution at present than it was twenty thousand or more years ago?
 - (a) The human genome has fully adapted to current environmental pressures
 - (b) Modern medicine has eliminated the effects of genetic drift
 - (c) The rate of genetic drift is directly proportional to technological advancements
 - (d) Humans have reached a genetic equilibrium where drift is no longer possible
 - (e) Humans have developed immunity to genetic drift

5. Chromosomal sex in mammals has been shown to be strongly Y chromosome determined. How was this shown to be true?
 - (a) The absence of a Y chromosome determines maleness in mammals
 - (b) The presence of a Y chromosome usually determines maleness in mammals
 - (c) The ratio of two X chromosomes to one Y chromosome determines maleness
 - (d) The presence of an X chromosome determines maleness in mammals
 - (e) The presence of an X chromosome in the absence of a Y chromosome determines maleness

True / False (10 marks)

Answer True or False

6. True or False: Oxidative phosphorylation primarily produces ATP, NADH, and FADH₂, rather than NADP⁺, ADP, and sugar.
7. True or False: A gal^+ / gal^- cell produced by abortive transduction will result in a colony that is minute when grown on minimal medium.
7. True or False: Proteins destined for the lysosome require two different signal peptides for their targeting.
8. True or False: The stages of a behavioral act typically include cue, desire, and fulfillment.
9. True or False: The acrosome and cortical reactions ensure that one and only one sperm fertilizes the egg, preventing twin zygotes from forming.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of a gene frequency of 0.07 for widow's peak, a dominant trait, by calculating and analyzing the expected frequencies of homozygous dominant, heterozygous, and homozygous recessive individuals in a population.
12. Discuss the concept of transgressive segregation in the context of genetics. How does it manifest in the F₂ generation, and what implications does it have for genetic variability?

End of Paper